

Model-Hamiltonian Replication of the Spin–Orbit Density Wave (SODW)

Hidden-Order Phenomenology in URu₂Si₂

TEXTURES-100 replication subagent
Paper: T. Das, *Phil. Mag.* (2014), arXiv:1406.5271

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Abstract

We reproduce, from scratch and at the model-Hamiltonian level, the central phenomenology of Das’s spin–orbit density wave (SODW) theory of the hidden-order (HO) phase of URu₂Si₂. The paper’s own numerics use WIEN2k DFT bands downfolded near E_F ; that material-specific input is deliberately *scoped out* here. Instead we build a transparent two-orbital, spin–orbit-split tight-binding band nested by $Q = (\pi, \pi)$, construct the paper’s SODW Nambu Hamiltonian (Eq. 2), and solve the interorbital mean-field gap equation (Eqs. 7, 10) self-consistently. The model yields a self-consistent SODW gap $\Delta_0 \simeq 6.2$ meV (paper: $\Delta_0 \sim 5$ –10 meV) and a transition temperature $T_h^{\text{model}} \simeq 17.9$ K, in close agreement with the experimental $T_h = 17.5$ K. The gap opens at E_F with a reconstructed DOS. Quantities that depend on the *fraction* of the Fermi surface removed — the entropy release and the spectral-weight loss — are under-reproduced by the thin model band, so the verdict is **PARTIAL**. Credit: the reusable `ollie_multipolar_stevens_landau_kernel`.

1 Paper claim

Das models the HO of URu₂Si₂ as a *spin–orbit density wave*: a particle–hole condensate at a nesting vector Q between two spin–orbit-split $5f$ bands, driven by interorbital Coulomb interaction V in the presence of SOC λ . Unlike an SDW it breaks translational symmetry while preserving time-reversal (no ordered moment), matching the “hidden” character. The headline: the SODW gap reproduces the HO gap phenomenology — FS gapping, $\sim 24\%$ entropy loss at $T_h = 17.5$ K, $\sim 40\%$ FS spectral-weight loss — and predicts a $T = 0$ critical field near the experimental $B_c \sim 35$ T.

2 Method (from scratch)

In the nested particle–hole subspace the SODW Bloch Hamiltonian reduces to

$$H_k = \begin{pmatrix} \varepsilon_1(k) & \Delta \\ \Delta^* & \varepsilon_2(k+Q) \end{pmatrix}, \quad E_{\pm}(k) = \varepsilon_{\pm}(k) \pm \sqrt{\varepsilon_{-}(k)^2 + |\Delta|^2}, \quad (1)$$

with $\varepsilon_{\pm} = \frac{1}{2}(\varepsilon_1(k) \pm \varepsilon_2(k+Q))$. The interorbital mean-field self-consistency (Das Eq. 10, ph channel) is the density-wave gap equation

$$1 = \frac{V}{N} \sum_k \frac{f(E_-) - f(E_+)}{2\sqrt{\varepsilon_{-}^2 + \Delta^2}}. \quad (2)$$

We use $\varepsilon_1 = -2t(\cos k_x + \cos k_y) + \delta/2$, $\varepsilon_2 = +2t(\dots) - \delta/2$, $t = 100$ meV ($W = 8t = 0.8$ eV), orbital split $\delta = 20$ meV, SOC scale $\lambda = 60$ meV, $Q = (\pi, \pi)$, on a 128×128 grid. We iterate the gap equation to convergence, sweep V (critical V) and T (transition T_h), and compute the Lorentzian DOS, the entropy $S(T) = -k_B \sum [f \ln f + (1-f) \ln(1-f)]/N$, and an order-of-magnitude Zeeman gap-closing field $2\Delta_0 = g\mu_B B_c$.

3 Results

Quantity	Model (this work)	Paper / experiment
SODW gap Δ_0	6.15 meV	$\sim 5\text{--}10$ meV
Transition T_h	17.9 K	17.5 K
$2\Delta_0/k_B T_h$	8.0	3.53 (BCS ref)
Critical interaction V_c	~ 0.30 eV	$V \sim 0.6$ eV (for $\Delta_0 \sim 10$)
FS spectral-weight loss	$\sim 6\%$	$\sim 40\%$
Entropy release	$\sim 0.006 k_B \ln 2$	$\sim 24\% R \ln 2$
Zeeman B_c	~ 106 T	~ 35 T

Table 1: Model vs. paper. Gap magnitude, T_h and V_c scale are reproduced; integrated-FS quantities (entropy, spectral-weight loss) and the field scale are not, because the single nested pocket of the model band gaps too little phase space.

4 Assessment

Verdict: PARTIAL. The self-consistent SODW mean field reproduces the two headline single-particle numbers — the gap magnitude in the 5–10 meV window and, strikingly, $T_h \approx 17.9$ K against the experimental 17.5 K — and the interaction scale $V_c \sim 0.3$ eV is the right order for $5f$ electrons. The *named gap* is in the extensive thermodynamic and spectral-weight fractions: our transparent model gaps only a small nested strip, so the entropy release and the $\sim 40\%$ FS spectral-weight loss come out roughly an order of magnitude too small, and the naive Zeeman B_c ($2\Delta_0/g\mu_B \approx 106$ T) overshoots the 35 T seen experimentally (the paper’s B_c folds in the g -factor and the field dependence of Δ , not captured by a static gap-closing estimate). These fractions are exactly the quantities the paper obtains from the realistic DFT Fermi surface, which was scoped out here. The qualitative SODW mechanism — an interorbital, SOC-enabled, time-reversal-preserving density wave that gaps E_F at a nesting Q — is fully reproduced.

Credit

Structural angular-momentum / Stevens-operator and Landau mean-field scaffolding informed by the reusable TEXTURES-100 kernel `ollie_multipolar_stevens_landau_kernel.py` (Ollie multipolar Stevens/Landau kernel). The SODW nesting construction, gap equation, DOS and entropy were built from scratch on top of that spirit.